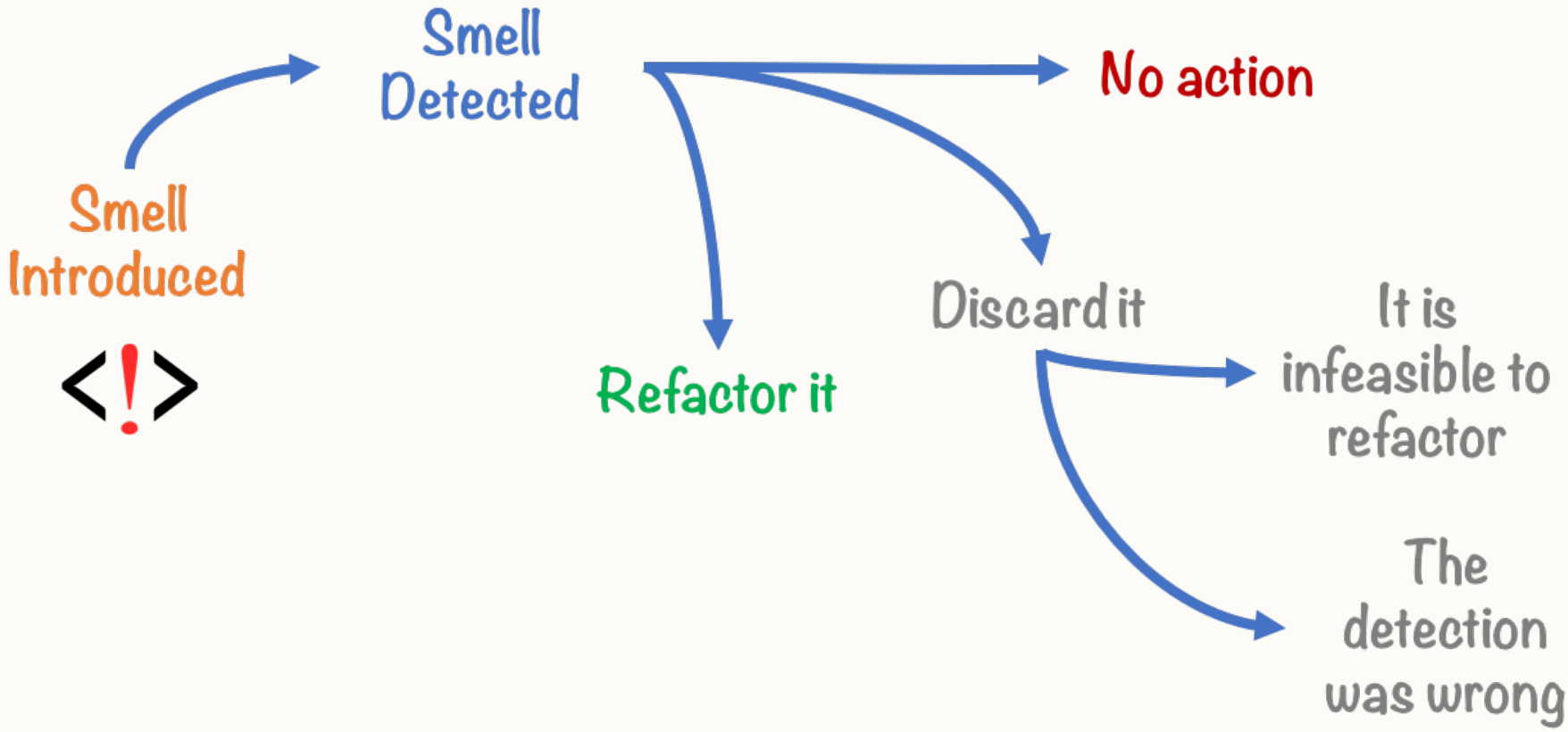


Identify Code Smells

Decide what you are going to refactor



What is Code Smell?





“A code smell is a surface indication that usually correspondence to a deeper problem in the system”

-Martin Fowler



What is Code Smell?

- Towards warning signs, indicating potential poor coding practices.

- **Not errors or bugs and do not impair the software's functionality**.
  - Reveal weaknesses in design that can slow down development or increase the risk of faults or failures further down the line.

- Code smells are inherently **subjective**.
  - One developer regards as a smell, but another might well see as a perfectly reasonable solution.
  - Decades of collective experience and often point to violations of fundamental software design principles.



Why code smells matters your opinions

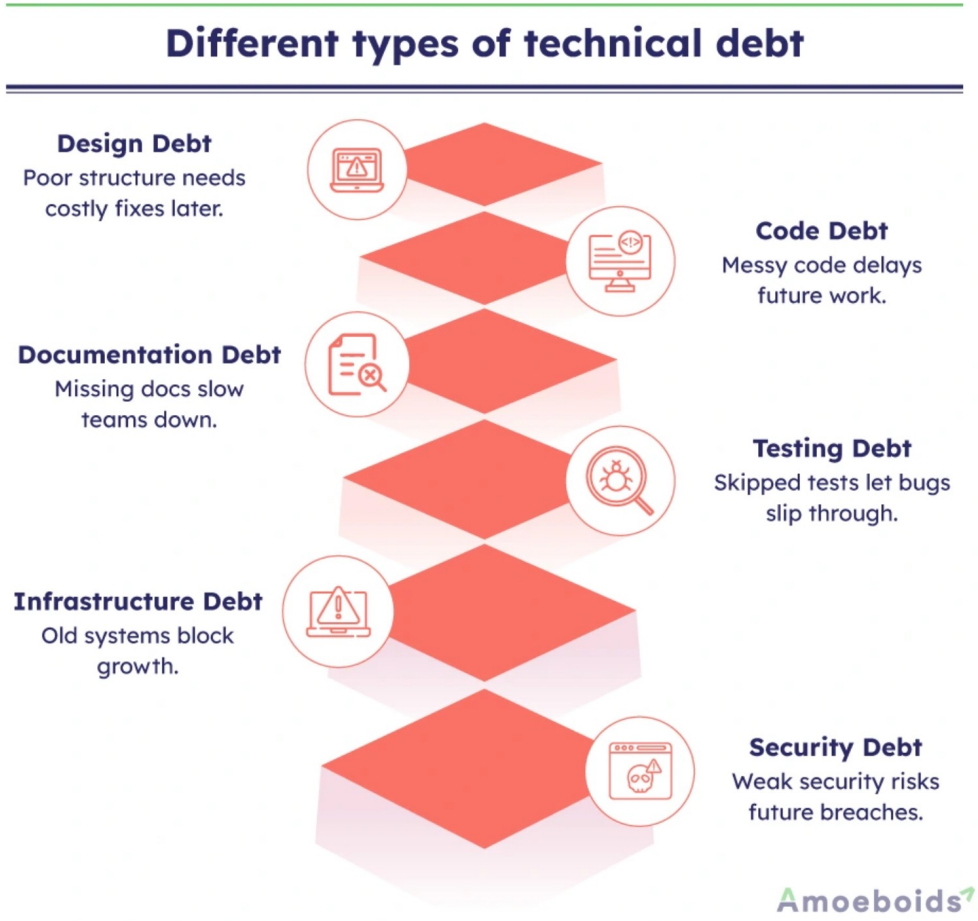
TEST



Why Code Smell Matters?

- They act as early warning signs that your code may be:
  - **Hard to understand**: It becomes difficult for other developers (or your future self) to read.
  - **Hard to change**: A simple change can require modifications in numerous places.
  - **Hard to test**: Tightly coupled code is almost impossible to unit-test properly, linking back to our legacy-code discussion.
  - **Bug-prone**: complex, tangled code is more likely to hide subtle faults.

Technical debt



https://amoeboids.com/blog/what-is-technical-debt/

Which type of technical debt most damaging

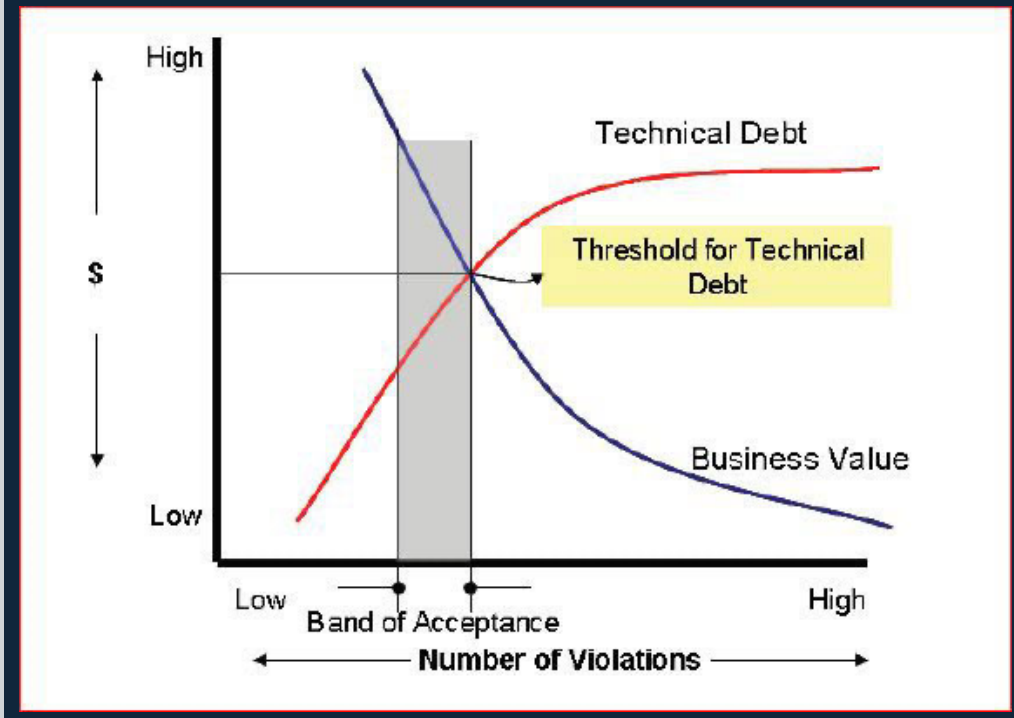
A. Architectural
B. Documentation
C. Code-level
D. Testing





Why Code Smell Matters?

- Technical debt compounds as developers compensate for smelly code without actually doing anything to eliminate it.





Code Smells vs Bugs

Item	Bugs	Code Smells
Definition	Flaws in code that cause incorrect behaviour, crashes or unexpected outputs.	Indicators of poor design that may lead to future bugs or maintenance issues.
Characteristics	•Cause visible errors during execution. •Can be reproduced with specific inputs. •Need immediate fixing. (High: Must fix) •Clearly right or wrong behaviour.	•Do not necessarily cause immediate issues. •Based on best practices and experience. •Addressed to avoid future problems. (Medium: Should fix) •Long-term code healthcare.



Code Smells vs Bugs

- Fix **BUGS** to make the software work correctly now.
- Address **CODE SMELLS** to make the software maintainable for the future.
- Code smells often create an environment where bugs are more likely to be introduced.
- Good developers fix bugs;
- Great developers also eliminate code smells that could cause future bugs



Code Smells Because Its Hard to Understand (not because it is incorrect)

- Duplicated Code:
  - bad because if you modify one instance of duplicated code but not the others, you (may) have introduced a bug!
- Long Methods:
  - long methods are more difficult to understand
  - Note: performance concerns with respect to lots of short methods are largely unfounded
- Large Classes:
  - classes try to do too much, which reduces cohesion (violates single responsibility, “The Blob”
- Long Parameter Lists:
  - hard to understand, can become inconsistent
- Divergent Changes:
  - related to cohesion where one type of change requires changing one subset of methods; another type of change requires changing another subset



More code smells examples

- Lazy Classes:
  - A class that no longer “pays its way”, part formed functionality
- Speculative Generality:
  - “Oh I think we need the ability to do this kind of thing someday”
- Temporary Field:
  - An attribute of an object is only set in certain circumstances; but an object should need all of its attributes
- Data Class

  - These are classes that have fields, getting and setting methods for the fields, and nothing else; they are data holders, but objects should be about data AND behavior
- Refused Bequest
  - A subclass ignores most of the functionality provided by its superclass where the subclass may not pass the “IS-A” test (Liskov Substitution Violation)
- Comment-based coverup
  - Comments are sometimes used to hide bad code, symptomatic of large blocks of comments, not using automatic comment generation



Still Confused? Try This:

- Many more smells at: http://c2.com/cgi/wiki?CodeSmell
- Taxonomy of code smells: http://www.tusharma.in/smells/
- An easy read: https://medium.com/@tusharma/how-to-track-code-smells-effectively-48dbf5ba659d